Assignment-2 - Solutions

① for, buck-boost, (CCM) $V_o = \frac{-V_{in} D}{1-D}$ $\Rightarrow$ if $V_o = -V_{in}$, then $\boxed{D = \frac{1}{2}}$

② $V_{out} = V_{in}$

③ ZERO

④ Flux Linkage

⑤ Decrease

⑥ $\boxed{I_{avg} = \frac{10}{\pi} = 3.1831\,A}$, $\boxed{I_{rms} = \frac{10}{2} = 5\,A}$

$$V_t I_{avg} + I_{rms}^2 r_d = 2.6465\,W$$

⑦ during t_o:-

power loss $= \frac{1}{2} \times t_o \times I_o \times V_K \times f_s = 0.0225\,W$

during t_a:-

power loss $= \frac{1}{2} \times t_a \times I_{RM} \times V_K \times f_s = 0.005625\,W$

during t_b:-

Power loss $= \frac{(V_R - 2V_K)\, I_{RM}\, t_b\, f_s}{6} = 0.663667\,W$

Power loss during $(t_a + t_b) = 0.669292\,W$.

Power loss during $(t_o + t_a + t_b) = 0.691792\,W$

(8) $t_{rr} = 5\mu s$, $S.F = \frac{t_b}{t_a} = 0.5$

$t_{rr} = t_a + t_b \Rightarrow t_{rr} = t_a\left[1 + \frac{t_b}{t_b}\right]$

$\therefore \boxed{t_a = \frac{5\mu}{1.5} = 3.3334\mu s.}$

$\boxed{t_b = 1.6667\mu s}$

$\therefore I_{RM} = t_a \frac{di}{dt} = 3.3334\mu \times 80\frac{A}{\mu s} = \underline{266.672A}$

$Q_{RR} = \frac{1}{2} t_{rr} . I_{RM} = \underline{666.68\mu C}$

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(10) $r(T_j) = \left[\frac{1.8R - R}{125 - 25}\right](T_j - 25) + R$

for r_1 $\underline{R = 1.8}$

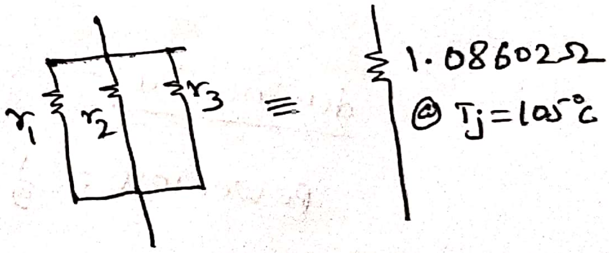
$r_1(T_j = 105°) = 2.952\Omega$

for r_2 , $\underline{R = 2}$

$r_2(T_j = 105°) = 3.28\Omega$

for r_3 , $\underline{R = 2.2}$

$r_3(T_j = 105°) = 3.608\Omega$



$I_{rms} = I_0\sqrt{D}$

$\boxed{I_{rms} = 3.535534A}$

$\because$ $\underline{50\% \text{ duty.}}$

power loss in r_1 = $\underline{4.994W}$

power loss in r_2 = $\underline{4.49481W}$

power loss in r_3 = $\underline{4.0862W}$

Total power loss = 13.575W.

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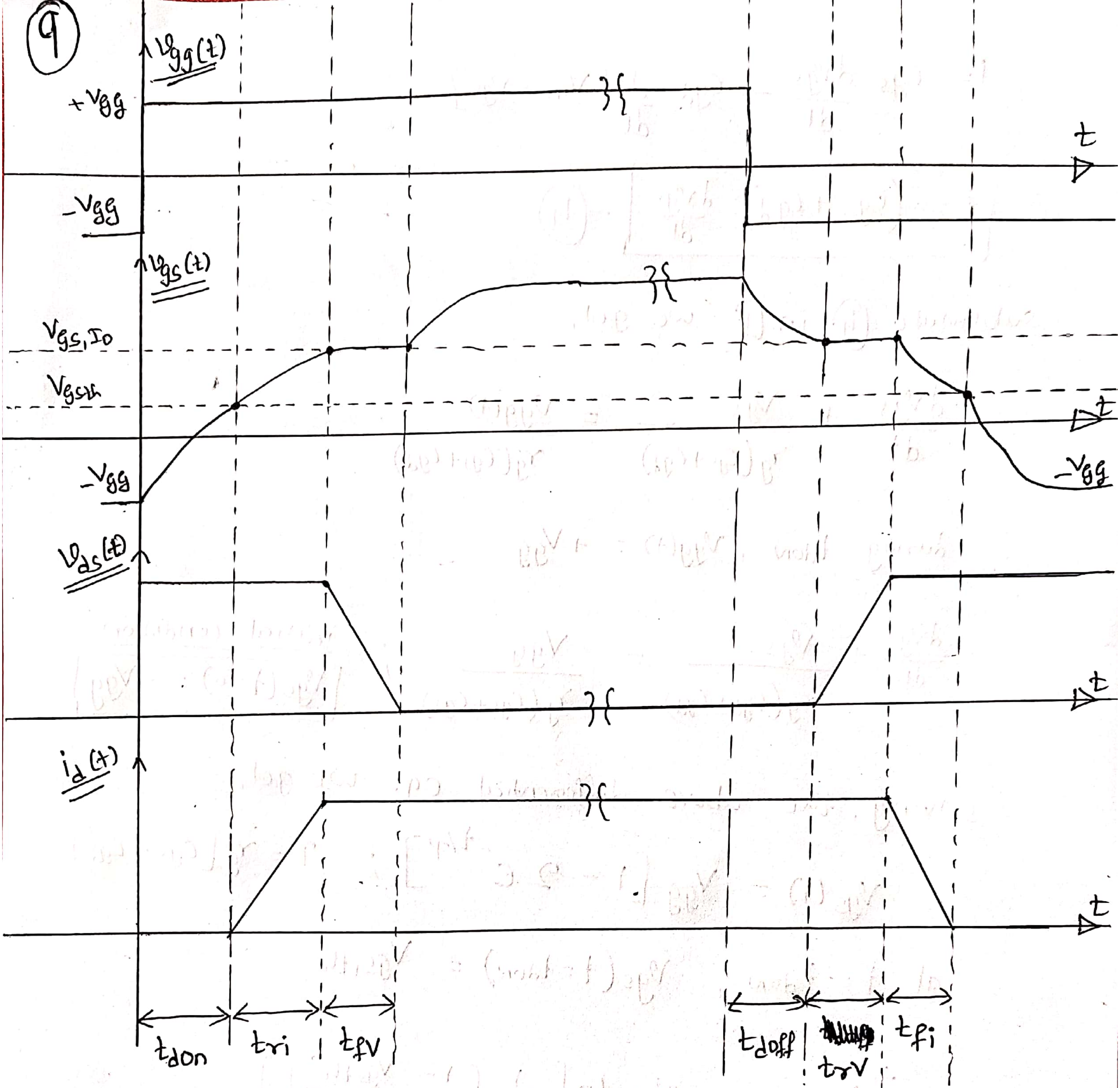
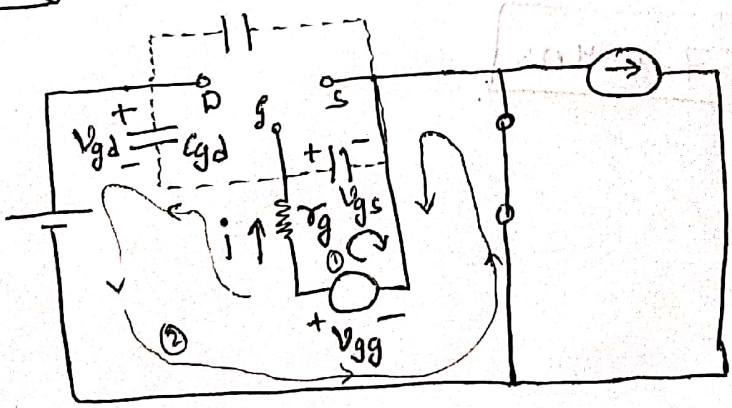


(a). During turn-on delay (t_{don}):-



$v_{gg}(t) - i r_g - v_{gs}(t) = 0$ — (1)

$v_{gg}(t) - i r_g + v_{gd}(t) - V_{in} = 0$ — (2)

from (1) & (2)

$\boxed{v_{gs} + v_{gd} = V_{in}}$ — (3)

now, $i = C_{gs}\frac{dv_{gs}}{dt} + \left[-C_{gd}\frac{dv_{gd}}{dt}\right]$

$$i = C_{gs}\frac{dv_{gs}}{dt} - C_{gd}\frac{d}{dt}[V_{in} - v_{gs}]$$

$$\boxed{i = (C_{gs} + C_{gd})\frac{dv_{gs}}{dt}} \quad -(4)$$

substitute (4) in (1) we get,

$$\frac{dv_{gs}}{dt} + \frac{v_{gs}}{r_g(C_{gs}+C_{gd})} = \frac{v_{gg}(t)}{r_g(C_{gs}+C_{gd})}$$

$\therefore$ during t_{don} ; $v_{gg}(t) = +V_{gg}$

$$\frac{dv_{gs}}{dt} + \frac{v_{gs}}{r_g(C_{gs}+C_{gd})} = \frac{V_{gg}}{r_g(C_{gs}+C_{gd})};$$

initial condition

$$\boxed{v_{gs}(t=0) = -V_{gg}}$$

solving, the above differential eq. we get,

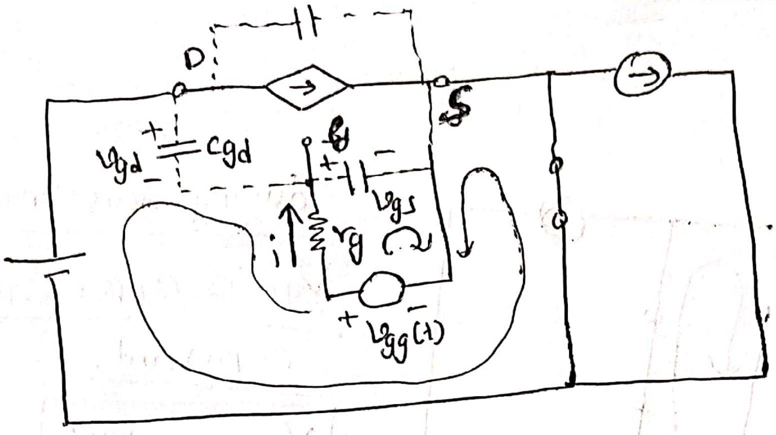
$$v_{gs}(t) = V_{gg}\left[1 - 2e^{-t/\tau}\right]; \quad \tau = r_g[C_{gs} + C_{gd}].$$

at $t = t_{don}$, $v_{gs}(t = t_{don}) = V_{gs,th}$

$$\therefore t_{don} = -\tau \times \ln\left[\frac{1}{2}\left[1 - \frac{V_{gs\,th}}{V_{gg}}\right]\right].$$

$$\therefore \boxed{t_{don} = 57.689\,ns}$$

(b) during current rise interval (t_{ri})



$$v_{gg}(t) - i r_g - v_{gs} = 0 \quad -(1)$$

$$v_{gg}(t) - i r_g + v_{gd} - V_{in} = 0 \quad -(2)$$

from (1) & (2)

$$\boxed{v_{gs} + v_{gd} = V_{in}} \quad -(3)$$

$$i = C_{gs}\frac{dv_{gs}}{dt} + \left[-C_{gd}\frac{dv_{gd}}{dt}\right]$$

$$\boxed{i = (C_{gs} + C_{gd})\frac{dv_{gs}}{dt}} \quad -(4)$$

Substitute (4) in (1) we get,

$$\frac{dv_{gs}}{dt} + \frac{v_{gs}}{r_g(C_{gs}+C_{gd})} = \frac{V_{gg}}{r_g(C_{gs}+C_{gd})}$$

In this interval initial condition

$v_{gs}(t=0) = V_{gs,th}$

Solution to above diff. eq. $\Rightarrow$ $v_{gs}(t) = V_{gg} + \left[\left[V_{gs,th} - V_{gg}\right] e^{-t/\tau}\right]$; $\tau = r_g(C_{gs}+C_{gd})$

@ $t = t_{ri}$ $\Rightarrow$ $v_{gs}(t) = V_{gs,Io}$.

$$\therefore t_{ri} = -\tau \times \ln\left[\frac{V_{gs,Io} - V_{gg}}{V_{gs,th} - V_{gg}}\right] = 18.31 ns$$

$$\therefore \boxed{t_{ri} = 18.31 ns}$$

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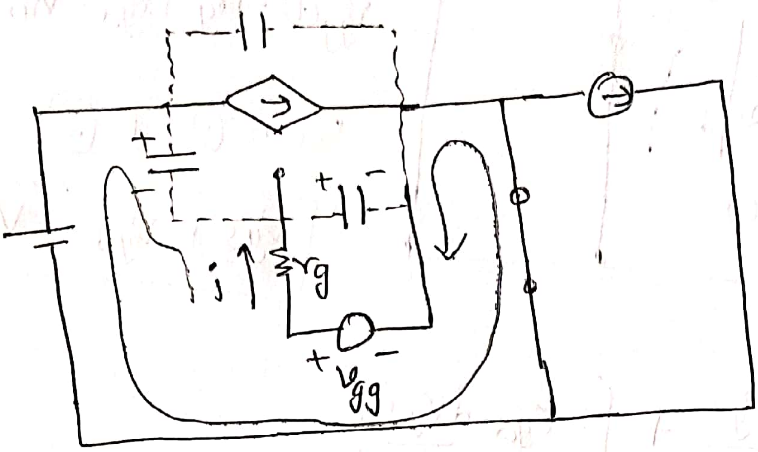
(c) during voltage fall interval (t_{fv}):-



during this interval V_{gs} is approximately constant.

$$\boxed{V_{gs,I_o} = 7V}$$

$\boxed{v_{gd} + v_{gs} = v_{ds}}$, as v_{gs} is constant $\boxed{\frac{dv_{gd}}{dt} = \frac{dv_{ds}}{dt}}$

$$-C_{gd}\frac{dv_{gd}}{dt} = \frac{V_{gg} - V_{gs,I_o}}{r_g}$$

$$\frac{dv_{ds}}{dt} = -\left[\frac{V_{gg} - V_{gs,I_o}}{r_g C_{gd}}\right]$$

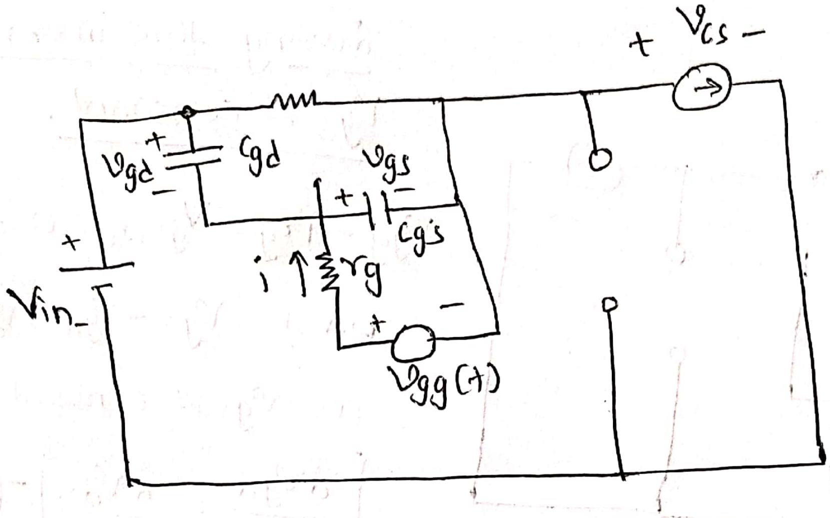
$$\therefore v_{ds}(t) = V_{in} - \left[\left[\frac{V_{gg} - V_{gs,I_o}}{r_g C_{gd}}\right] t\right]$$

$\therefore$ @ $t = t_{fv} \Rightarrow v_d \approx 0$

$$\therefore t_{fv} = \frac{V_{in} \times r_g \times C_{gd}}{V_{gg} - V_{gs,I_o}} = 281.25\,ns.$$

$$\therefore \boxed{t_{fv} = 281.25\,ns}$$

ⓓ during turn-off delay. (t_{doff}):- $\boxed{v_{gg}(t) = -V_{gg}}$



$v_{gg}(t) - i r_g - v_{gs} = 0$ -(1)

$v_{gg}(t) - i r_g + v_{gd} - V_{in} - V_{cs} = 0$ -(2)

$V_{cs} = V_{in} - I_o r_{ds on}$.

approximately

$\boxed{V_{cs} \approx V_{in}}$

$\therefore$ eq.(2) $\rightarrow$ $v_{gg}(t) - i r_g + v_{gd} = 0$ -(3) $\Rightarrow$ from (1) & (3) $\boxed{v_{gs} = -v_{gd}}$

$$i = C_{gs}\frac{dv_{gs}}{dt} + \left[-C_{gd}\frac{dv_{gd}}{dt}\right] \Rightarrow i = (C_{gs}+C_{gd})\frac{dv_{gs}}{dt} \quad -(4)$$

substituting (4) in (1) $\Rightarrow$ $\frac{dv_{gs}}{dt} + \frac{v_{gs}}{r_g(C_{gs}+C_{gd})} = \frac{-V_{gg}}{r_g(C_{gs}+C_{gd})}$

solving the differential equation, we get,

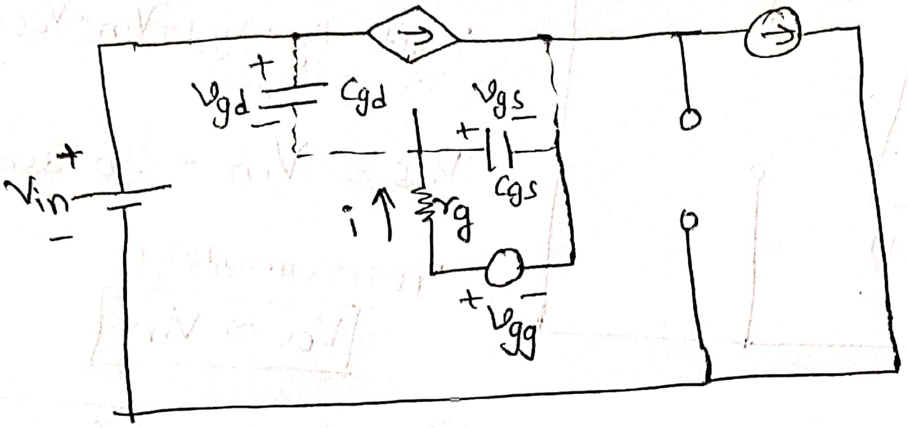
$$v_{gs} = -V_{gg} + 2V_{gg}\, e^{-t/\tau} \; ; \quad \tau = r_g(C_{gs}+C_{gd}).$$

$$\therefore t_{doff} = -\tau \ln\left[\frac{V_{gs,I_o} + V_{gg}}{2V_{gg}}\right]$$

$\because$ at $t = t_{doff}$ $\Downarrow$ $v_{gs}(t) = V_{gs,I_o}$

$$\therefore \boxed{t_{doff} = 17.833\,ns}$$

(e) during voltage rise interval (t_{rv}):-



during this interval $v_{gs} \approx$ constant.

$$v_{gg} - i r_g - V_{gs,I_o} = 0 \quad \text{—(1)}$$

w.k.t. $v_{gs} + v_{gd} = v_{ds}$ —(2)

$\therefore$ as $v_{gs} \approx$ constant

$$\boxed{\frac{dv_{gd}}{dt} = \frac{dv_{ds}}{dt}} \quad \text{—(3)}$$

$$\therefore i = \frac{v_{gg}(t) - V_{gs,I_o}}{r_g} = \frac{-V_{gg} - V_{gs,I_o}}{r_g}$$

$$-C_{gd}\frac{dv_{gd}}{dt} = -\left[\frac{V_{gg} + V_{gs,I_o}}{r_g}\right], \quad \text{as } \frac{dv_{gd}}{dt} = \frac{dv_{ds}}{dt}$$

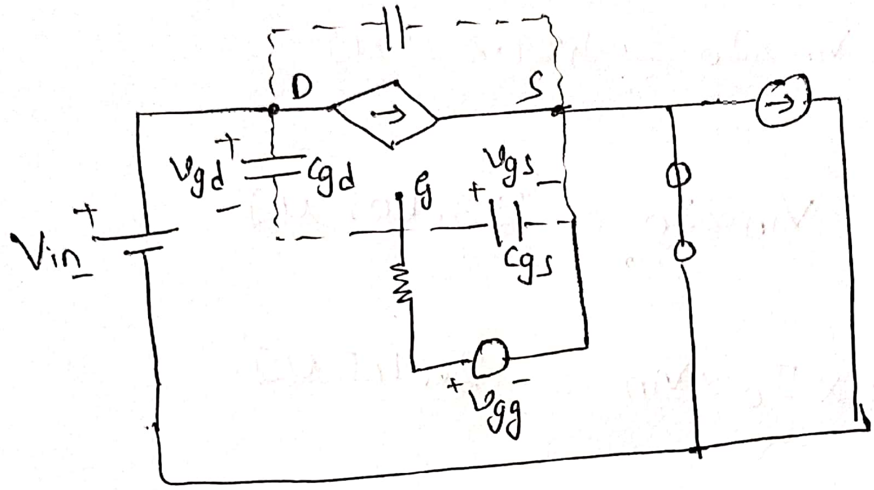
$$\frac{dV_{ds}}{dt} = \frac{V_{gg} + V_{gs,I_o}}{r_g C_{gd}} \Rightarrow \boxed{V_{ds} = \frac{(V_{gg} + V_{gs,I_o})}{r_g C_{gd}} \cdot t}$$

at the end of this interval, $V_{ds} \approx V_{in}$.

$$\therefore t_{rv} = \left[\frac{V_{in} \cdot r_g \cdot C_{gd}}{V_{gg} + V_{gs,I_o}}\right]$$

$$\therefore \boxed{t_{rv} = 102.27 ns}$$

(f) During current fall interval (t_{fi}):-



$$v_{gg}(t) - i r_g - v_{gs} = 0 \quad -(1)$$

$$v_{gg} - i r_g - v_{gd} - V_{in} = 0 \quad -(2)$$

from (1) & (2) $\boxed{v_{gs} + v_{gd} = V_{in}}$

$$i = C_{gs}\frac{dv_{gs}}{dt} + \left[-C_{gd}\frac{dv_{gd}}{dt}\right] = (C_{gs}+C_{gd})\frac{dv_{gs}}{dt} \quad -(3)$$

Substitute (3) in (1),

$$\frac{dv_{gs}}{dt} + \frac{v_{gs}}{r_g(C_{gs}+C_{gd})} = \frac{-V_{gg}}{r_g(C_{gs}+C_{gd})}$$

@ $t=0$
$v_{gs} = V_{gs,I_o}$

solution to the above diff. eq. is,

$$v_{gs} = -V_{gg} + \left[V_{gs,I_o} + V_{gg}\right]e^{-t/\tau}; \quad \tau = r_g(C_{gs}+C_{gd})$$

$$\therefore t_{fi} = -\tau \ln\left[\frac{V_{gs,th} + V_{gg}}{V_{gs,I_o} + V_{gg}}\right]$$

@ $t = t_{fi}$
$v_{gs} = V_{gs,th}$

$$\therefore \boxed{t_{fi} = 8.43\,ns}$$

(a) $E_{ri} = \frac{1}{2} \times t_{ri} \times I_o \times V_{in} = 27.34 \mu J$

(b) $E_{fv} = \frac{1}{2} \times t_{fv} \times V_{in} \times I_o = 421.875 \mu J$

(c) $E_{rv} = \frac{1}{2} \times t_{rv} \times V_{in} \times I_o = 153.405 \mu J$

(d) $E_{fi} = \frac{1}{2} \times t_{fi} \times I_o \times V_{in} = 12.645 \mu J$

Energy lost during the MOSFET ON time.

(e) $E_{ON} = I_o \times (I_o \times r_{dson}) \times \left[(0.5T_s) - (t_{don} + t_{ri} + t_{fv}) + t_{doff} \right]$

$\therefore \boxed{E_{ON} = 1.23303 \text{ mJ}}$

$\therefore$ Total power loss in mosfet $= \left[E_{ri} + E_{fv} + E_{rv} + E_{fi} + E_{ON} \right] \times f_s$

$= 36.97 W.$

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